

UNIT TWO DERIVATIVE IN USE

Section 2.1 Elasticity: Definition and as Logarithmic Derivative

Elasticity is a measure of a proportionate change in the dependent variable which results from a proportionate change in the independent variable. Thus elasticity of demand is a measure of the percentage change in quantity demanded due to a percentage change in the factors which determine demand.

If there is change in quantity demanded from Q to $(Q + \Delta Q)$ resulting from a given change in price from P to $(P + \Delta P)$, the price elasticity of demand is

$$\begin{aligned} E_p &= \frac{\text{Proportionate change in quantity demanded}}{\text{Proportionate change in price}} \\ &= \frac{\frac{\Delta Q}{Q}}{\frac{\Delta p}{p}} \\ &= \frac{\Delta Q}{Q} \times \frac{p}{\Delta p} = \frac{\Delta Q}{\Delta p} \left(\frac{p}{Q} \right) \end{aligned}$$

It gives us the average, price elasticity of demand over the price range of P to $(P + \Delta p)$. It is possible to use the method of derivative to get the price elasticity of demand at a particular point P , i.e., as Δp approaches to zero.

$$\begin{aligned} E_p &= \lim_{\Delta p \rightarrow 0} \frac{p}{Q} \left(\frac{\Delta Q}{\Delta p} \right) = \frac{P}{Q} \lim_{\Delta p \rightarrow 0} \frac{\Delta Q}{\Delta p} \\ &= \frac{P}{Q} \left(\frac{dQ}{dp} \right) \end{aligned}$$

Thus price elasticity of demand

$$E_p = \frac{dQ}{dp} \left(\frac{p}{Q} \right)$$

Example

1. Find the price elasticity of demand if the demand function for a product is $Q = P^2 - 6P + 36$, where Q is quantity and P is price of a product. Using the above formula

$$E_p = \frac{dQ}{dp} \left(\frac{p}{Q} \right), \text{ but } \frac{dQ}{dp} = 2p - 6$$

$$\text{Therefore, } E_p = (2p - 6) \frac{p}{P^2 - 6P + 36}$$

$$E_p = \frac{2p^2 - 6p}{p^2 - 6p + 36}$$

$$E_p = \frac{2p^2 - 6p}{p^2 - 6p + 36}$$

What is the elasticity of demand at $P = 4$?

$$E_p = \frac{2(4^2) - 6(4)}{4^2 - 6(4) + 36}$$

$$= \frac{32 - 24}{16 - 24 + 36} = \frac{8}{28} < 1$$

As $E_p < 1$ demand is said to be price inelastic at $P = 4$.

2. Find the price elasticity of demand in terms of quantity sold.

$$P = \frac{10}{(Q+1)^2} = 10 (Q+1)^{-2}$$

$$E_p = \frac{dQ}{dp} \left(\frac{P}{Q} \right), \text{ Therefore, } \frac{1}{E_p} = \frac{dp}{dQ} \left(\frac{Q}{P} \right)$$

Differentiating the demand function with respect to Q gives as

$$\frac{dp}{dQ} = -20 (Q+1)^{-3} = \frac{-20}{(Q+1)^3}$$

$$\text{Thus, } \frac{1}{E_p} = \left[\frac{-20}{(Q+1)^3} \right] \frac{Q}{\frac{10}{(Q+1)^2}}$$

$$\frac{1}{E_p} = \frac{-2Q}{(Q+1)}$$

This means

$$E_p = \frac{Q+1}{-2Q}$$

3. If the demand function is given as $Q = \sqrt{10 - p^2}$, find the price elasticity of demand at $P = 2$.

$$E_p = \frac{dQ}{dp} \frac{p}{Q} \quad \frac{dQ}{dp} = \frac{1}{2} (10 - p^2)^{-1/2} (-2p)$$

$$= \frac{-2p}{2(10 - p^2)^{1/2}}$$

$$\frac{dQ}{dp} = \frac{-p}{\sqrt{10 - p^2}}$$

$$\text{Thus, } E_p = \left(\frac{-p}{\sqrt{10 - p^2}} \right) \left(\frac{p}{\sqrt{10 - p^2}} \right)$$

$$E_p = \frac{-p^2}{10-p^2} \cdot \text{Thus, at } P=2, E_p = \frac{-2^2}{10-2^2} = \frac{-4}{6} = \frac{-2}{3}$$

This means, as the price of the product increases by one percent then the quantity demanded for the product decreases by $\frac{2}{3}$ percent at $p = 2$.

Elasticity as Logarithmic Derivative

In a function $y = f(t)$, when the dependent variable y is a function of time t , then the instantaneous rate of growth is determined as

$$\text{Rate of growth} = \frac{dy/dt}{y} = \frac{\text{marginal function}}{\text{total function}}$$

This ratio exactly represents the derivative of $\ln f(t)$. This means, the instantaneous rate of growth of the function is the derivative of natural logarithm of the function with respect to time. That is

$$\text{Rate of growth} = \frac{d \ln^{f(t)}}{dt}$$

By now, given the function $y = f(x)$, let us observe the result provided that we carry out differentiation of $\ln^y$ with respect to $(\ln^x)$. To start it, suppose $u = \ln^y$ and $v = \ln^x$. Then the chain of relationship relating u to y and v to x is that

$$u = \ln^y, y = f(x), U = \ln^x, x = e^v$$

Finding the derivative of $\ln y$ with respect to $\ln x$ gives us

$$\frac{d(\ln^y)}{d(\ln^x)} = \left(\frac{du}{dy}\right) \left(\frac{dy}{dx}\right) \left(\frac{dx}{dv}\right)$$

Using the method of substitution

$$\begin{aligned} \frac{d(\ln^y)}{d(\ln^x)} &= \left[\frac{d(\ln^y)}{dy} \right] \left(\frac{dy}{dx} \right) \left[\frac{d(e^v)}{dv} \right] \\ &= \left(\frac{1}{y} \right) \frac{dy}{dx} e^v \\ &= \left(\frac{1}{y} \right) \frac{dy}{dx} x \text{ as } e^v = x \end{aligned}$$

$$\boxed{\frac{d(\ln^y)}{d(\ln^x)} = \frac{dy}{dx} \left(\frac{x}{y} \right)}$$

As you remember this expression is the point elasticity of the function. Hence, one can generalize that for a function $y = f(x)$, the point elasticity of y with respect to x is

$$E_{xy} = \frac{d(\ln y)}{d(\ln^x)}$$

Example

1. Find the point elasticity of demand given $Q = K/P^n$ where K and n are positive constants

$$Q = K p^{-n}$$

$$\ln Q = \ln K - n \ln P$$

$$Ep = \frac{d(\ln Q)}{d(\ln p)} = -n$$

$$|Ep| = |-n| = n$$

2. Find the point elasticity of demand provided that $x = a p^b$ where a and b are positive constants, and x and p are quantity demanded and price of a product respectively. $\ln x = \ln a + b \ln p$

$$\text{Thus, } Ep = \frac{d(\ln x)}{d(\ln p)} = b$$

Relationship between Marginal revenue, average revenue and elasticity of Demand

Given the total revenue function $TR = PQ$ where P is the price at which the product is sold and Q is the quantity sold. Then marginal revenue that is the rate of change of total revenue with respect output is

$$MR = \frac{d(PxQ)}{dQ}$$

Using the product rule of differentiation

$$MR = p \frac{dQ}{dQ} + Q \frac{dp}{dQ}$$

$$MR = P + \frac{Q dp}{dQ}$$

We know from the above analysis that $Ep = \frac{dQ}{dp} \frac{p}{Q}$ which implies that $\frac{1}{Ep} = \frac{dp}{dQ} \left(\frac{Q}{p}\right)$

multiplying both sides by p gives us $\frac{p}{Ep} = \left(\frac{dp}{dQ}\right) \left(\frac{Q}{p}\right) p$

$$\frac{p}{Ep} = Q \frac{dp}{dQ}$$

Substituting it the above marginal revenue function give us

$$MR = P + \frac{P}{Ep}$$

However, we also know that p is equal to average revenue (AR), thus

$$MR = AR + \frac{AR}{Ep}$$

$$MR = AR \left(1 + \frac{1}{Ep}\right) = AR \left(1 - \frac{1}{|Ep|}\right)$$

$$MR = AR \left(1 - \frac{1}{|Ep|}\right)$$

From this equation we realize that

When demand is unitary elastic, i.e. $|Ep| = 1$, $MR = 0$ this means, total revenue is maximized.

When demand is elastic, i.e., $|E_p| > 1$, $MR > 0$ which implies that total revenue is increasing. However, as demand is inelastic, i.e., $|E_p| < 1$, $MR < 0$ which implies that total revenue is decreasing.

Section 2.2 Higher Order Derivatives

We know that the first order derivative $f'(x)$ of a function $y = f(x)$ is a function of x . As a result we can determine the rate of change of $f'(x)$ with respect to x if $f'(x)$ is differentiable. The result of this differentiation is referred to as second order derivative of the function $y = f(x)$. It is represented by $f''(x)$ where the double prime shows that the primitive function has been differentiated with respect to x twice. The expression (x) next to the double prime indicates that the second order derivative is a function of x .

Alternatively this second order derivative can be represented by

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$$

Dear colleague! We have seen that the second derivative is a function of x . Thus, it is possible to differentiate this function with respect to x in order to get the third order derivative of the function $f'''(x)$ or $f^3(x)$ which in turn can be a source of fourth order derivative and so on provided that the differentiable condition is satisfied.

Symbolically, these successive higher order derivatives are

$$f^3(x), f^4(x), f^5(x), \dots, f^n(x).$$

Or

$$\frac{d^3y}{dx^3}, \frac{d^4y}{dx^4}, \frac{d^5y}{dx^5}, \dots, \frac{d^n(x)}{dx^n}$$

Example

1. Find the first through the fourth derivative of the following function.

$$f(x) = 5x^3 + 2x^2 + 3x + 4$$

These derivatives are $f'(x) = 15x^2 + 4x + 3$

$$f''(x) = 30x + 4$$

$$f'''(x) = 30$$

$$f^4(x) = 0$$

We have seen that each successive derivative gives us a simpler expression than its precedence until we get the fourth derivative. However, it is not always true. The fourth order derivative is equal to zero does not imply that it does not exist.

2. Find the first four derivatives of the function given as

$$f(x) = \frac{2x}{1-x}, \quad x \neq 1$$

$$f'(x) = \frac{2(1-x) - (-1)(2x)}{(1-x)^2}, \text{ using the quotient rule}$$

$$f'(x) = \frac{2(1-x) - (-1)(2x)}{(1-x)^2} = \frac{2}{(1-x)^2}$$

$$f''(x) = 2(-2)(1-x)^{-3}(-1)$$

$$f''(x) = -4(-1)(1-x)^{-3} \text{ using the chain rule}$$

$$= 4(1-x)^{-3}$$

$$f^3(x) = (-3)(4)(1-x)^{-4}(-1)$$

$$f^3(x) = 12(1-x)^{-4}$$

$$f^4(x) = (-4)(12)(1-x)^{-4}(-1)$$

$$f^4(x) = 48(1-x)^{-4}$$

2.2.1 Convexity and Concavity of a Function

Given the function $y = f(x)$, it is clear that the derivative of the function $f'(x)$ indicates the rate of change of the primitive function with respect to x . Similarly, $f''(x)$ shows the rate of change of $f'(x)$ with respect to x . Observing the sign of the first derivative of the original function at any point of x , one can determine whether a function is increasing or decreasing. However, the sign of the second derivative indicates whether the function is concave or convex at that particular point. If the original function is $y = f(x)$,

- $f'(a) > 0$ implies that the function is increasing at $x = a$ and
- $f'(a) < 0$ implies that the function is decreasing at $x = a$
- $f'(a) = 0$, implies that the function is at its optimum point at $x = a$

Similarly

- If $f''(a) > 0$, then the function is convex at $x = a$
- If $f''(a) < 0$, then the function is concave at $x = a$
- If $f''(a) = 0$, $x = a$ is the point at which the curvature of the function is changed.

In general, a positive first derivative coupled with a positive second derivative at $x = a$ shows that the function increases at an increasing rate at $x = a$. A positive first derivative coupled with a negative second derivative at $x = a$, indicates that the original function is increasing at a decreasing rate at this point.

A negative first derivative with a positive second derivative shows that the function is decreasing at an increasing rate where as a negative first derivative with negative second derivative at $x = a$, implies that the function decreases at a decreasing rate at $x = a$.

Example

If $f(x) = x^3 - 12x^2 + 36x + 8$,

- a) Is it increasing or decreasing at $x = 3$?
- b) Is it convex or concave at $x = 3$?

Solution

First we should find the first and second derivative of this function at $x = 3$ to answer these questions.

$$f'(x) = 3x^2 - 24x + 36$$

$$f''(x) = 6x - 24$$

$$\text{Therefore, } f'(3) = 3(9) - 24(3) + 36$$

$$f'(3) = 27 - 72 + 36 = 63 - 72 = -9 < 0$$

$$f''(3) = 6(3) - 24 = 18 - 24 = -6 < 0$$

This implies that the function is decreasing at decreasing rate at $x = 3$ and it is concave at this point.

Linear Approximation

As we know there are complicated functions in economics. In order to avoid this complexity, we sometimes try to find a simpler function which approximates the original function. Given the function $y = f(x)$, the equation of a straight line which passes through point (x_1, y_1) and having a slope b is represented as

$$b = \frac{y - y_1}{x - x_1}$$

$$b(x - x_1) = (y - y_1)$$

$$y = b(x - x_1) + y_1$$

This means, the equation of the tangent line which passes through the graph of the function $y = f(x)$ at $x = a$ provided that $f(x)$ is continuous and smooth at $x = a$, is expressed as

$$y - f(a) = f'(a)(x - a)$$

$$y = f'(a)(x - a) + f(a)$$

When we approximate the graph of $f(x)$ by its tangent line at $x = a$, then the resulting approximation is referred to as linear approximation.

Example

1. Find the linear approximation of the function $f(x) = \sqrt[4]{x}$, about $x = 1$.

First we should determine the value of $f(x)$ and $f'(x)$ at $x = 1$ to answer this question. Thus,

$$f'(x) = \frac{1}{4} x^{\left(\frac{1}{4}-1\right)} = \frac{1}{4} x^{\left(-\frac{3}{4}\right)}$$

$$f'(x) = \frac{1}{4} (1)^{-\frac{3}{4}} = \frac{1}{4}$$

$$f(1) = \sqrt[4]{1} = 1$$

Therefore, the linear approximation of the function is

$$f(x) \approx \frac{1}{4} (x-1) + 1$$

$$\begin{aligned}
 \text{If } x = 1.02, f(x) &\approx \frac{1}{4} (1.02 - 1) + 1 \\
 &\approx \frac{1}{4} (0.02) + 1 \\
 f(x) &\approx \frac{1}{200} + 1 = 1.005
 \end{aligned}$$

However, the actual value is $f(1.02) = \sqrt[4]{1.02} = 1.00496$ which is very close to the approximate value of the function, i.e., 1.005.

2. Determine the linear approximation of the function

$$f(x) = (1 + \frac{3}{2}x + \frac{1}{2}x^2)^{\frac{1}{2}} \quad \text{at } x = 0$$

$$f'(x) = \frac{1}{2} (1 + \frac{3}{2}x + \frac{1}{2}x^2)^{-\frac{1}{2}} (\frac{3}{2} + x)$$

This means $f'(0) = \frac{1}{2} (1 + \frac{3}{2}(0) + \frac{1}{2}(0))^{\frac{1}{2}} (\frac{3}{2} + 0)$

$$f'(0) = \frac{1}{2} (1)^{-\frac{1}{2}} (\frac{3}{2}) = \frac{1}{2} (\frac{3}{2}) = \frac{3}{4}$$

And $f(0) = [1 + \frac{3}{2}(0) + \frac{1}{2}(0)]^{\frac{1}{2}}$

$$f(0) = (1)^{\frac{1}{2}} = 1$$

Therefore, at $x = 0$ the linear approximation of the function is

$$f(x) \approx f'(0)(x - 0) + f(0)$$

$$f(x) \approx \frac{3}{4}(x - 0) + 1$$

$$f(x) \approx \frac{3}{4}x + 1$$

3. Identify an approximate value of $(1.001)^{50}$

Solution

In order to approximate it, initially we should convert the equation in to

$f(x) = x^{50}$, when x is close to 1. Then we should evaluate the original function and the first derivative of this function at $x=1$.

Then $f'(x) = 50x^{49}$

$$f'(1) = 50(1)^{49} = 50$$

And $f(1) = (1)^{50} = 1$

Thus, $f(x) \approx f(1) + f'(1)(x-1)$

$$\approx 1 + 50(x-1) = 1 + 50x - 50 = 50x - 49$$

$$(1.001)^{50} \approx 1 + 50(1.001 - 1)$$

$$\approx 1 + 50(0.001) = 1 + 0.05 = 1.05$$

However, the actual value is $(1.001)^{50} = 1.0512$ which is very close to the approximate value i.e., 1.05

2.2.3. Polynomial Approximation

Approximation using linear functions is not as such accurate. Therefore, it is necessary to use quadrate approximation or higher order approximation minimizes this problem.

a) Quadratic Approximation

Given a function $y = f(x)$, how can it be approximated by the second degree polynomial of the form $P(a) = A + B(x - a) + C(x - a)^2$ as x is close to a ?

In this function we have seen three unknown coefficients such as A , B and C . Therefore, we ought to put the following three conditions in order to determine these three unknowns.

At $x = a$, it is assumed that

- $f(a) = P(a)$
- $f'(a) = P'(a)$
- $f''(a) = P''(a)$

Given the polynomial function $P(x) = A + B(x - a) + C(x - a)^2$,

$$P'(x) = B + 2C(x - a)$$

$$P''(x) = 2C$$

$$\text{When } x = a, \quad P(a) = A + B(a - a) + C(a - a)^2 \\ = A = f(a)$$

$$P'(a) = B + 2C(a - a)$$

$$P'(a) = B = f'(a)$$

$$P''(a) = 2C = f''(a), \quad C = \frac{1}{2} f''(a)$$

Substituting these values in the given quadratic function gives us the quadratic approximation of the function which is represented by

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{1}{2} f''(a)(x - a)^2$$

as x is close to a .

Example

1 Determine the quadratic approximation to a function $f(x) = \sqrt[4]{x}$ about $x = 1$. In this case, first we should find the value of $f(x)$, $f'(x)$ and $f''(x)$ at $x = 1$. Thus,

$$f(x) = x^{1/4}, \quad f(1) = 1^{1/4} = 1$$

$$f'(x) = \frac{1}{4} x^{-3/4}, \quad f'(1) = \frac{1}{4} (1)^{-3/4} = \frac{1}{4}$$

$$f''(x) = \frac{1}{4} \left(\frac{-3}{4} \right) x^{-7/4}, \quad f''(1) = \frac{-3}{16}$$

Thus,

$$f(X) \approx f(1) + f'(1)(x - 1) + \frac{1}{2} f''(1)(x - 1)^2 \\ \approx 1 + \frac{1}{4}(x - 1) + \frac{1}{2} \left(\frac{-3}{16} \right) (x - 1)^2$$

$$f(x) \approx 1 + \frac{1}{4}(x-1) - \frac{3}{32}(x-1)^2$$

Compare the actual value and the approximate value of this function when $x = 1.02$.

2. Find the quadratic approximation to a function $f(x) = (5x + 3)^{-2}$ about $x = 0$.

Solution

$$f(0) = \frac{1}{9}$$

$$f'(x) = -2(5x+3)^{-3}(5) = -10(5x+3)^{-3}$$

$$f'(0) = -10[5(0)+3]^{-3} = -10(3)^{-3} = \frac{-10}{27}$$

$$f''(x) = 30(5x+3)^{-4}(5) = 150(5x+3)^{-4}$$

$$f''(0) = 150(5(0)+3)^{-4} = 150(3)^{-4} = \frac{150}{81}$$

Therefore, the quadratic approximation of the function is

$$\begin{aligned} f(x) &\approx f(0) + f'(0)(x-0) + \frac{1}{2}f''(0)(x-0)^2 \\ &\approx \frac{1}{9} + \left(-\frac{10}{27}\right)(x-0) + \frac{1}{2}\left(\frac{150}{81}\right)(x-0)^2 \\ f(x) &\approx \frac{1}{9} - \frac{10}{27}x + \frac{75}{81}x^2 \end{aligned}$$

B. Higher Order Approximation

Quadratic approximation is not as such accurate. Therefore, functions with higher order derivatives can be better approximated near one point by using polynomial function of higher degree.

Given a function $y = f(x)$, how can we approximate it around $x = a$ by an n^{th} degree polynomial of the form

$$P(x) = A_0 + A_1(x-a) + A_2(x-a)^2 + A_3(x-a)^3 + \dots + A_n(x-a)^n$$

As we have seen this function has $(n+1)$ unknown coefficients. Therefore, we should impose $(n+1)$ conditions on the polynomial to determine the value of the unknown coefficients. These are, at $x = a$, it is assumed that

$$f(a) = P(a)$$

$$f'(a) = P'(a)$$

$$f''(a) = P''(a)$$

⋮

$$f^{(n)}(a) = P^{(n)}(a)$$

Given the above polynomial function,

$$P'(x) = A_1 + 2A_2(x-a) + 3A_3(x-a)^2 + 4A_4(x-a)^3 + \dots + nA_n(x-a)^{n-1}$$

$$P''(x) = 2A_2 + 6A_3(x-a) + 12A_4(x-a)^2 + \dots + n(n-1)A_n(x-a)^{n-2}$$

$$P'''(x) = 6 A_3 + 24 A_4 (x - a) + \dots + n(n-1)(n-2) A_n (x-a)^{n-3}$$

,

$$P^n(x) = n(n-1)(n-2)(n-3) \dots (3)(2) A_n$$

Therefore, when x is close to a,

$$P(a) = A_0 = f(a)$$

$$P'(a) = 1! A_1 = f'(a), A_1 = f'(a)/1!$$

$$P''(a) = 2! A_2 = f''(a), A_2 = f''(a)/2!$$

$$P^3(a) = 3! A_3 = f^3(a), A_3 = f^3(a)/3!$$

,

$$P^n(a) = n! A_n = f^n(a), A_n = f^n(a)/n!$$

Substituting these values of the unknown coefficients in the polynomial function gives us the polynomial approximation of the function as x is close to a, which is represented by

$$f(x) \approx f(a) + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \frac{f^3(a)(x-a)^3}{3!} + \dots + \frac{f^n(a)}{n!} (x-a)^n$$

Example

Find the fourth order approximation of the function

$$f(x) = \sqrt{1+x}, \text{ about } x = 0$$

$$\text{In this case } f(x) \approx f(0) + f'(0)(x-0) + \frac{f''(0)}{2!} (x-0)^2 + \frac{f^3(0)}{3!} (x-0)^3 + \frac{f^4(0)}{4!} (x-0)^4$$

$$f'(x) = \frac{1}{2}(1+x)^{-1/2}, f'(0) = \frac{1}{2}(1+0)^{-1/2} = \frac{1}{2}$$

$$f''(x) = \frac{-1}{4}(1+x)^{-3/2}, f''(0) = -\frac{1}{4}$$

$$f^3(x) = \frac{3}{8}(1+x)^{-5/2}, f^3(0) = \frac{3}{8}$$

$$f^4(x) = \frac{-15}{16}(1+x)^{-7/2}, f^4(0) = \frac{-15}{16}$$

Therefore,

$$\sqrt{1+x} \approx 1 + \frac{1}{2}(x-0) + \frac{(-1/4)(1/2)}{2!} (x-0)^2 + \frac{3}{8} \left(\frac{1}{3!}\right) (x-0)^3 - \frac{15}{16} \left(\frac{1}{4!}\right) (x-0)^4$$

$$\sqrt{1+x} \approx 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{3}{48}x^3 - \frac{15}{384}x^4$$

$$\sqrt{1+x} \approx 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \frac{15}{384}x^4$$

2.2.4 Estimation of Functions (Maclaurin and Taylor series)

A. Maclaurin series (Expansion of a function around $x = 0$)

It is an expansion of a function $y = f(x)$ that is differentiable, around $x = 0$. In other words, it represents the polynomial approximation of the function around $x = 0$. Therefore, given the function

$$f(x) = A_0 + A_1x + A_2x^2 + A_3x^3 + \dots + A_n x^n$$

Repeatedly differentiating this function gives us

$$f'(x) = A_1 + 2A_2x + 3A_3x^2 + \dots + n A_n x^{n-1}$$

$$f''(x) = 2A_2 + 6A_3x + \dots + n(n-1) A_n x^{n-2}$$

$$f^{(3)}(x) = 6A_3 + 24 A_4 x + \dots + n(n-1)(n-2) A_n x^{n-3}$$

$$f^{(n)}(x) = n(n-1)(n-2)(n-3) \dots (3)(2)(1) (A_n), (n = \text{Positive integer})$$

What do understand from these successive derivatives? We have understood that the number of constant terms is decreased by one until the n^{th} derivative is reached. We can determine the value of each rate of change at various values of x . In this case, let us evaluate these derivatives at $x = 0$. When we determine the value of the derivatives at $x = 0$, then all terms containing x will be eliminated.

$$f'(0) = A_1, f''(0) = 2A_2, f^{(3)}(0) = 3(2) A_3, f^{(4)}(0) = (4)(3)(2) A_4 \dots$$

$$f^{(n)}(0) = n(n-1)(n-2)(n-3) \dots (3)(2)(1) A_n$$

Using the symbol $(n!)$ which can be read as n – factorial $(n!)$, where $n! = n(n-1)(n-2)(n-3) \dots (3)(2)(1)$

$$A_1 = \frac{f'(0)}{1!}, A_2 = \frac{f''(0)}{2!}, A_3 = \frac{f^{(3)}(0)}{3!}, \dots, A_n = \frac{f^{(n)}(0)}{n!}$$

Now the primitive function $f(x)$ can be expressed as a new polynomial function by substituting these terms in it as follows

$$f(x) = \frac{f(0)}{0!} + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \frac{f^{(3)}(0)}{3!} x^3 + \dots + \frac{f^{(n)}(0)}{n!} x^n$$

In this case, the unknown coefficients of the primitive function are replaced by the derivatives evaluated at $x = 0$ or and $0!$ is equal to 1.

This power series representation is known as Maclaurin series of the primitive function $f(x)$ around $x = 0$

Example

1. Determine the Maclaurin series of the function

$$f(x) = 5x^3 + 2x^2 + 3x + 1$$

This function has the following derivatives

$$f'(x) = 15x^2 + 4x + 3$$

$$f''(x) = 30x + 4$$

$$f^3(x) = 30$$

$$\text{So that } f(0) = 1$$

$$f'(0) = 15(0)^2 + 4(0) + 3 = 3$$

$$f''(0) = 30(0) + 4 = 4$$

$$f^3(0) = 30$$

Therefore, the Maclauris series of this function is

$$\begin{aligned} f(x) &= f(0) + \frac{f'(0)}{1!}(x-0) + \frac{f''(0)}{2!}(x-0)^2 + \frac{f^3(0)}{3!}(x-0)^3 \\ &= 1 + 3(x-0) + \frac{4}{2}(x-0)^2 + \frac{30}{6}(x-0)^3 \\ f(x) &= 1 + 3x + 2x^2 + 5x^3 \end{aligned}$$

This result shows that the Maclaurin series really represents the given function $f(x)$.

2. Find the Maclaurin series of the function

$$f(x) = 3x^2 + 2x + 4$$

The derivatives from this function

$$f'(x) = 6x + 2$$

$$f''(x) = 6$$

$$\text{So that } f(0) = 4, f'(0) = 2, f''(0) = 6$$

$$\begin{aligned} \text{Thus, } f(x) &= f(0) + f'(0)(x-0) + \frac{f''(0)}{2!}(x-0)^2 \\ &= 4 + 2(x-0) + \frac{6}{2}(x-0)^2 = 4 + 2x + 3x^2 \end{aligned}$$

B. Taylor's Series

It deals about the expansion of the primitive function $f(x)$ provided that the function is continuous and smooth, around any point $x = x_0$. Thus given the function $y = f(x)$, we will explain it using a specific quadratic function for the sake of simplicity.

Let us consider the expansion of this function at some point $x = x_0$

$$f(x) = A_0 + A_1x + A_2x^2 + A_3x^3 + A_4x^4 + \dots + A_nx^n$$

First let us write this function in terms of the power of $(x-x_0)$

$$f(x) = A_0 + A_1(x-x_0) + A_2(x-x_0)^2 + A_3(x-x_0)^3 + \dots + A_n(x-x_0)^n$$

This function has the following successive derivatives

$$f'(x) = A_1 + 2A_2(x-x_0) + 3A_3(x-x_0)^2 + 4A_4(x-x_0)^3 + \dots + nA_n(x-x_0)^{n-1}$$

$$f''(x) = 2(1)A_2 + 3(2)(1)A_3(x-x_0) + \dots + n(n-1)A_n(x-x_0)^{n-2}$$

$$f^3(x) = 3(2)(1)A_3 + 4(3)(2)(1)A_4(x-x_0) + \dots + n(n-1)(n-2)A_n(x-x_0)^{n-3}$$

$$\vdots$$

$$f^n(x) = n(n-1)(n-2)(n-3)\dots(3)(2)(1)A_n$$

So that

$$\begin{aligned} f(x_0) &= 0! A_0, & A_0 &= \frac{f(x_0)}{0!} \\ f'(x_0) &= 1! A_1, & A_1 &= \frac{f'(x_0)}{1!} \\ f''(x_0) &= 2! A_2, & A_2 &= \frac{f''(x_0)}{2!} \\ &\vdots \\ f^n(x_0) &= n! A_n, & A_n &= \frac{f^n(x_0)}{n!} \end{aligned}$$

Therefore

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \dots + \frac{f^n(x_0)}{n!}(x-x_0)^n$$

This power series representation is referred to as Taylor's series. If we consider 3 as a point of expansion, then

$$f(x) = f(3) + f'(3)(x-3) + \frac{f''(3)}{2!}(x-3)^2 + \dots + \frac{f^n(3)}{n!}(x-3)^n$$

Example

Find the Taylor's series of the function $f(x) = 3x^2 + 2x + 5$, around $x = x_0$

The successive derivatives of this function are

$$f'(x) = 6x + 2$$

$$f''(x) = 6$$

So that $f(x_0) = 3x_0^2 + 2x_0 + 5$

$$f'(x_0) = 6x_0 + 2$$

$$f''(x_0) = 6$$

Thus, $f(x) = (3x_0^2 + 2x_0 + 5) + (6x_0 + 2)(x - x_0) + \frac{6}{2}(x - x_0)^2$

$$= (3x_0^2 + 2x_0 + 5) + (6x_0x - 6x_0^2) + 2x - 2x_0 + 3(x^2 - 2xx_0 + x_0^2)$$

$$= 5 - 6x_0^2 + 2x + 3x^2 + 6x_0^2$$

$$f(x) = 3x^2 + 2x + 5$$

This realizes that the Taylor's series correctly represents the given function

2.2.5 Intermediate value theorem and Newton Method

A. Intermediate Value Theorem

Suppose that there is a function $f(x)$ that is continuous for all values of x in the close interval $[a, b]$, and assume that the values of the function at $x = a$ and at $x = b$ are not equal to each other and each value has opposite sign, i.e., $f(a)$ and $f(b)$ have different signs. The intermediate value theorem states that there is at least one value of x say c in the interval $[a, b]$ such that $f(c) = 0$.

Example

1. Prove that the equation $x^6 + 3x^2 - 2x - 1 = 0$ has at least one solution in between 0 and 1, i.e. $[0, 1]$

You ought to follow the steps below to solve this Problem.

- First put the equation in a polynomial form of $f(x) = x^6 + 3x^2 - 2x - 1$
- Second check whether the function is continuous in between 0 and 1 or not.
- Third determine the value of the function at $x = 0$ and $x = 1$

As we know the function $f(x)$ is referred to as continuous in the interval (a, b) iff

i) $f(x)$ is continuous at $x = a$ [i.e., $\lim_{x \rightarrow a^+} f(x) = f(a)$]

ii) $f(x)$ is continuous at $x = b$ [i.e. $\lim_{x \rightarrow b^-} f(x) = f(b)$]

iii) $f(x)$ is continuous at $x = c$ [i.e., $a < c < b$]

As a given function in our example is a polynomial function, it is continuous in between 0 and 1.

$$f(0) = 0^6 + 3(0)^2 - 2(0) - 1 = -1$$

$$f(1) = 1^6 + 3(1)^2 - 2(1) - 1 = 1 + 3 - 2 - 1 = 1$$

We have seen that $f(0)$ and $f(1)$ have different signs. Therefore, according to the intermediate value theorem there is at least one number that is $c \in [0, 1]$ so that $f(c) = 0$. In other words, the above equation has at least one solution in between 0 and 1.

2. Prove that the equation $\sqrt{2x^2 + 3x} = 2$ has at least one solution in between 0 and 1, i.e. $[0, 1]$

First make it in to polynomial function by squaring both sides of the equation as

$$(\sqrt{2x^2 + 3x})^2 = 2^2$$

$$2x^2 + 3x = 4$$

$$2x^2 + 3x - 4 = 0, \text{ then } f(x) = 2x^2 + 3x - 4$$

As the function is polynomial it is continuous in between 0 and 1.

$$f(0) = 2(0)^2 + 3(0) - 4 = -4, \quad f(1) = 2 + 3 - 4 = 1$$

The value of this function has different signs at $x = 0$ and $x = 1$ which implies that there is at least one solution for the equation $\sqrt{2x^2 + 3x} = 2$ has at least one solution in $[0, 1]$.

3. Show that the equation $\sqrt{x^2 - 1} = 3x$ has at least one solution in between 0 and 1.

Solution

Initially change the equation in to polynomial function by squaring both sides as

$$(\sqrt{x^2 - 1})^2 = (3x)^2$$

$$-8x^2 + 1 = 0 \text{ thus, } f(x) = -8x^2 + 1$$

As the function is a polynomial function, it is continuous in between 0 and 1.

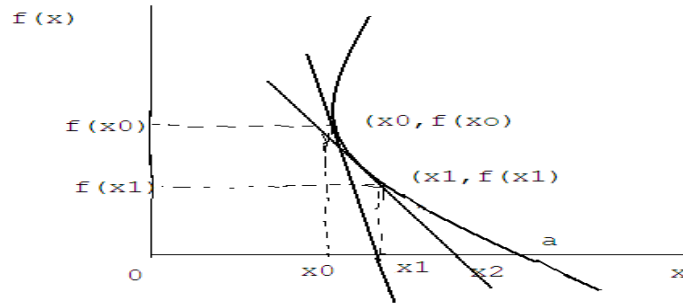
$$f(0) = 1 \text{ and } f(1) = -8(1)^2 + 1 = -7$$

We have observed that $f(0)$ and $f(1)$ have different signs. As a result, the equation has at least one solution in between 0 and 1.

B. Newton's Method

The intermediate value theorem does not provide information concerning the location at which the solution resides. Rather Newton's method leads to a better approximate solution of the equation under consideration.

Given the function $y = f(x)$ and assume $f(x) = 0$ at $x = a$. Estimate a . To carry out this activity, let us begin with an initial estimate x_0 of a . Constructing a tangent line which passes through $[x_0, f(x_0)]$ we get x_1 near to a as shown in the following figure. Undertaking the same procedure we



get the estimate x_1 by drawing a tangent line at $[x_1, f(x_1)]$. Continuing this process enables us to estimate the value of a . However, we can also easily find the formula for x_n derived by the Newton's method.

The slope of this line can be calculated as

$$m = \frac{y - y_0}{x - x_0} = \frac{f(x) - f(x_0)}{x - x_0}$$

$$f(x) - f(x_0) = m(x - x_0)$$

$$f(x) - f(x_0) = f'(x_0)(x - x_0)$$

When $x = x_1$, $f(x) = 0$ (for the tangent line passing through $[x_0, f(x_0)]$)

$$\text{Therefore, } 0 - f(x_0) = f'(x_0)(x_1 - x_0)$$

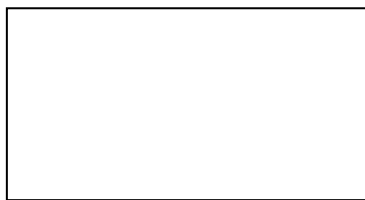
Rearranging this equation gives as

$$\frac{-f(x_0)}{f'(x_0)} = (x_1 - x_0)$$

This implies that $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

Similarly $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$, $x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$

In general the points generated by the Newton method are obtained by



Example

1. Find the approximate values of the equation $f(x) = x^6 + 3x^2 - 2x - 1$ in the interval $[0,1]$ using the Newton's method once.

Solution

In this case $n = 0$, and $x_0 = 1$

We know that $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

$$f(x_0) = f(1) = 1 + 3 - 2 - 1 = 1$$

$$f'(x_0) = 6x^5 + 6x - 2 \Rightarrow f'(x_0) = f'(1) = 6 + 6 - 2 = 10$$

$$\text{Therefore, } x_1 = 1 - \frac{1}{10} = 0.9$$

$$\begin{aligned} \text{If we check for } f(x_1) &= f(0.9) \\ &= (0.9)^6 + 3(0.9)^2 - 2(0.9) - 1 \\ &= 0.53441 + 2.43 - 1.8 - 1 \\ f(0.9) &= 0.161441 \end{aligned}$$

Applying Newton's method twice

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 0.9 - \frac{f(0.9)}{f'(0.9)}$$

$$= 0.9 - \frac{0.161441}{6.94294}$$

$$x_2 = 0.877$$

Thus, $f(0.877) = (0.877)^6 + 3(0.877)^2 - 2(0.877) - 1$
 $= 0.0084$ which is more approximate to zero.

2. Determine the approximate value of the equation $\sqrt{x^2 - 3x} = 3x$ in the interval $[0, 1]$ using Newton's method once.

Solution

Squaring both sides of the equation gives us $f(x) = -8x^2 + 1$. In this case also $n = 0$, $x_0 = 1$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$f(x_0) = (1)^2 (-8) + 1 = -7$$

$$f'(x_0) = f'(1) = -16(1) = -16$$

Therefore

$$x_1 = 1 - \frac{(-7)}{(-16)}$$

$$= 1 - 0.4375$$

$$x_1 = 0.5625 \approx 0.56$$

If we check the original function at this value of x , we get $f(0.56) = -8(0.56)^2 + 1$
 $= -2.5088 + 1$
 $f(0.56) = -1.5088$

Applying the Newton method twice gives us

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 0.56 - \frac{f(.56)}{f'(.56)}$$

$$x_2 = 0.56 - \frac{(-1.5088)}{(-8.96)} = 0.392$$

Then $f(0.392) = -8(0.392)^2 + 1 = -0.2293$ which is more approximate to the solution than the earlier value. This shows that we can determine the most approximate value of the solution by applying this procedure continuously.

Section 2.3 Multivariate Calculus

Some functions involve one dependent and more than one independent variable. These functions are said to be multivariate functions. For example, the production function $Q = f(K, L)$ has one dependent and two independent variables, labor L and capital K .

2.3.1 Partial Derivatives

Partial differentiation is a technique of deriving the rate of change of the function with respect to change in one of the independent variable when all other variables in the function are held constant. Thus, if the production function $Q = f(K, L)$ is differentiated with respect to L , with K

being constant, we obtain the rate of change of total product with respect to labor, i.e., Marginal product of labor (MP_L).

Partial differentiation follows the normal rules of differentiation apart from the fact that all variables other than the one the function is being differentiated with respect to are assumed to be constant.

Given the function, $y = f(x_1, x_2)$ the usual notation for the partial derivative of the function with respect to x_1 is

$$\frac{\partial y}{\partial x_1} = f_{x_1} \text{ and that of with respect to } x_2 \text{ is } \frac{\partial y}{\partial x_2} = f_{x_2}$$

Example

1. If the function is given by $y = 4x_1^2x_2^3 + 6x_1^2x_2 + x_1^4 + 7x_2^3$, find f_{x_1} and f_{x_2} .

$$f_{x_1} = 8x_1x_2^3 + 12x_1x_2 + 4x_1^3 \quad (x_2 \text{ is assumed to be constant})$$

$$f_{x_2} = 12x_1^2x_2^2 + 6x_1^2 + 21x_2^2 \quad (x_1 \text{ is assumed to be constant})$$

Economic Applications

Partial differentiation is frequently applied in economic analysis as the economy is a complex system to understand. Economists usually observe the effect of changes in one variable supposing all other variables constant.

A. Elasticity

As we know quantity demanded for a product depends on various factors such as price of the good, (p) consumer's income (y), price of related goods, and population (n).

$$Q_{dd} = f(p, y, p_c, p_s, n) \text{ where } p_c \text{ is price of complement.}$$

p_s is price of substitutes.

And we also know that price elasticity of demand E_p is

$$E_p = \frac{dQ}{dp} \left(\frac{p}{Q} \right)$$

But with the existence of these determinants in the demand function

$$E_p = \frac{\partial Q}{\partial P} \left(\frac{P}{Q} \right)$$

Example

Given the demand function as $Q = 35 - 0.4P + 0.15y - 0.25P_c + 0.12P_s + 0.003n$, what is the price elasticity of demand when price is 24?

$$E_p = \frac{\partial Q}{\partial P} \left(\frac{P}{Q} \right), \text{ you are required to get } \frac{\partial Q}{\partial P}.$$

$$\frac{\partial Q}{\partial P} = -0.4 \text{ (all factors other than the price of the product are constant)}$$

$$E_p = \frac{(-0.4)(24)}{35 - 0.4(24) + 0.15y - 0.25P_c + 0.12P_s + 0.003n}$$

$$\text{Thus, } E_p = \frac{(-9.6)}{25.4 + 0.15y - 0.25P_c + 0.12P_s + 0.003n}$$

The exact price elasticity of demand can be calculated if the values of the remaining variables are given.

B. Utility Functions

According to the Cardinality approach the consumer who consumes only two products A and B will have the utility function

$$U = U(A, B)$$

We can determine the marginal utility of each good from this function using the technique of partial differentiation.

Example

If the utility function of the consumer is given by

$U = 2A^{0.4} B^{0.4}$ Where A and B represent the quantity of the two goods consumed; find marginal utility of each good.

$$\begin{aligned} MU_A &= 2(0.4) A^{-0.6} B^{0.4} \\ &= \frac{0.8B^{0.4}}{A^{0.6}} \end{aligned}$$

$$\begin{aligned} MU_B &= 0.4(2) A^{0.4} B^{-0.6} \\ &= \frac{0.8A^{0.4}}{B^{0.6}} \end{aligned}$$

As the consumption of good A increases, MU_A decrease and as the consumption of B increases MU_B decreases. As a result the law of diminishing marginal utility holds for both goods.

C) Production Function

When the firm produces a product using two or more variable inputs, then we can determine the rate of change of total product with respect the change in each input using the method of partial differentiation.

Example

If the production function for a product is given by $Q = 20K^{0.5} L^{0.5}$ where K is capital and L is labor, Find the marginal Product of each input.

$$\begin{aligned} \text{Marginal product of labor (MP}_L\text{)} &= 0.5(20) K^{0.5} L^{-0.5} \\ MP_L &= \frac{10K^{0.5}}{L^{0.5}} \end{aligned}$$

$$\begin{aligned} \text{Marginal Product of capital (MP}_K\text{)} &= 0.5(20) K^{-0.5} L^{0.5} \\ MP_K &= \frac{10L^{0.5}}{K^{0.5}} \end{aligned}$$

We have seen that the law diminishing marginal productivity is satisfied for each input.

D) Revenue and Cost Functions

It is clear that some cases firms may produce different products. If they use common production facilities, the costs of each product will be related. Therefore, the marginal cost function of the individual product can be determined by the method of partial differentiation.

Example

A firm produces two products Q_1 and Q_2 and its total cost function is $C(Q_1, Q_2) = 20(Q_1 + Q_2)$. What are the relevant marginal cost functions.

$$C = 20Q_1 + 20Q_2$$

$$\text{Thus, Marginal cost of } Q_1 = \frac{\partial C}{\partial Q_1}$$

$$MC_1 = 20$$

$$\text{Marginal cost of } Q_2 (MC_2) = 20$$

In other cases, the firm may produce different products which are complementary with each other. This means, the price of one good will affect the quantity demanded of the other good. Thus, the marginal revenue of one good is the partial derivative of total revenue with respect to that good assuming the price of the other good constant.

Example

Suppose a firm produces two products A and B which are complements. The relevant demand functions are

$$Q_A = 850 - 12.5 P_A - 3.8 P_B$$

$$Q_B = 936 - 4.8 P_A + 24 P_B$$

Marginal revenue is usually expressed in terms of quantity. Thus, we should initially rewrite the demand functions in the form of price in terms of quantity.

$$\text{For good A, } Q_A = 850 - 12.5 P_A - 3.8 P_B$$

$$12.5 P_A = 850 - 3.8 P_B - Q_A$$

$$P_A = \frac{850 - 3.8 P_B - Q_A}{12.5}$$

$$TRA = P_A Q_A = \frac{850 Q_A - 3.8 P_B Q_A - Q_A^2}{12.5}$$

$$\text{Therefore MRA} = \frac{\partial TRA}{\partial Q_A}$$

$$MR_A = 68 - 0.304 P_B - 0.16 Q_A$$

$$\text{For good B, } Q_B = 936 - 4.8 P_A - 24 P_B$$

$$24 P_B = 936 - 4.8 P_A - Q_B$$

$$P_B = 39 - 0.2 P_A - \frac{Q_B}{24}$$

$$\text{But } TR_B = P_B Q_B = 39 Q_B - 0.2 P_A Q_B - \frac{Q_B^2}{24}$$

$$MR_B = \frac{\partial TR_B}{\partial Q_B}$$

$$MR_B = 39 - 0.2 P_A - \frac{1}{12} Q_B$$

The marginal revenue functions of the above two goods show that when the demand for the two goods are interrelated, the marginal revenue function for one good will depend on the price level of the other good.

E) Keynesian Multiplier

If we incorporate the government sector and foreign trade, the basic Keynesian macroeconomic model will be

$$Y = C + I + G + (X - M)$$

and the functional relationship with consumption is

$$C = c Y_d$$

Where c is the marginal Propensity to consume, and imports

$$M = m Y_d$$

Where m is marginal propensity to import,

$$Y_d = (1-t) Y$$

is the disposable income c , m and t are parameters and Investment I , Government expenditure G and export X are exogenously determined.

Therefore $Y = c Y_d + I + G + [X - (m Y_d)]$

$$Y = c (1-t) Y + I + G + X - m (1-t) Y$$

$$Y - c (1-t) Y + m (1-t) Y = I + G + X$$

$$Y [1 - c (1-t) + m (1-t)] = I + G + X$$

$$Y = \frac{I + G + X}{1 - c(1-t) + m(1-t)} = \frac{I + G + x}{1 - (c - m)(1-t)}$$

If we are interested to determine the government expenditure multiplier, we should partially differentiate income with respect to G , holding Investment I and export X constant as shown below.

$$\frac{\partial Y}{\partial G} = \frac{1}{1 - (c - m)(1-t)}$$

$$\text{However, } \frac{\partial Y}{\partial G} = \frac{\partial Y}{\partial X} = \frac{\partial Y}{\partial I}$$

Example

In a Keynesian Macroeconomic model of an economy

$$Y = C + I + G + (X - M) \quad I = 820$$

$$\begin{aligned}
C &= 0.75 Y_d & G &= 960 \\
M &= 0.25 Y_d & Y_d &= (1-t) Y \\
X &= 650 & t &= 0.3
\end{aligned}$$

Determine the equilibrium level of national income. Using the export multiplier find out what will happen to the balance of trade if export exogenously increases by 100.

Initially, we should solve the relationship between income (Y) with consumption and Import.

$$\begin{aligned}
C &= 0.75 Y_d = 0.75 (1-t) Y = 0.75(1-0.3) Y \\
&= 0.75 (0.7) Y \\
C &= 0.525 Y \\
M &= 0.25 Y_d = 0.25 (1-0.3) Y \\
&= 0.25 (0.7) Y \\
M &= 0.175 Y
\end{aligned}$$

Therefore, the equilibrium level of national income is

$$\begin{aligned}
Y &= 0.525Y + 820 + 960 + 650 - 0.175Y \\
Y - 0.525Y + 0.175Y &= 2430 \\
0.65Y &= 2430 \\
Y &= \frac{2430}{0.65} = 3,738.46
\end{aligned}$$

At this equilibrium level of income, import will be

$$\begin{aligned}
M &= 0.25 (0.7) (3,738.46) \\
M &= 654.23
\end{aligned}$$

Thus, the balance of payment will be

$$\begin{aligned}
X - M &= 650 - 654.23 \\
&= -4.23 \text{ which is negative trade balance.}
\end{aligned}$$

However, as export increases by 100, then the equilibrium level of income will be increased by change in export times export multiplier, i.e.,

$$\begin{aligned}
\Delta Y &= \Delta X \frac{\partial Y}{\partial X} = 100 \left[\frac{1}{1 - (c - m)(1 - t)} \right] \\
&= 100 \left[\frac{1}{1 - (0.75 - 0.25)(1 - 0.3)} \right] \\
&= 100 \left[\frac{1}{1 - (0.5)(0.7)} \right] \\
&= 100 \left[\frac{1}{0.65} \right]
\end{aligned}$$

$$\Delta Y = 153.85$$

As a result, equilibrium level of income increases from 3, 738. 46 to 3, 892.31 when export increases from 650 to 750. At this new level of equilibrium income, import will be

$$\begin{aligned}
M &= 0.175 (3,892.31) \\
M &= 681.15
\end{aligned}$$

And the new balance of trade will be

$$\begin{aligned}
X - M &= 750 - 681.15 \\
&= 68.85
\end{aligned}$$

Therefore, the balance of trade changes from 4.23 deficits to t8.85 surplus when export exogenously increases by 100.

2.3.2 Second Partial and Young's Theorem

As you know, if the first order partial derivatives of the function are continuous and differentiable, differentiating these functions will result in second order partial derivatives.

Given the function $Z = f(x, y)$, if it is differentiable, it has two first order partial derivatives such as $Z_x = \frac{\partial Z}{\partial x}$ and $Z_y = \frac{\partial Z}{\partial y}$. In addition to these, this function has four second order partial derivatives (direct and cross partial derivatives). These are

$$Z_{xx} = \frac{\partial^2 Z}{\partial x^2}$$

$$Z_{yy} = \frac{\partial^2 Z}{\partial y^2}$$

These two derivatives are referred to as direct second order partial derivatives where as

$$Z_{xy} = \frac{\partial Z_y}{\partial x} \text{ and } Z_{yx} = \frac{\partial Z_x}{\partial y} \text{ are said to be second order cross partial}$$

derivatives. According to the young's theorem, the mixed (cross) partial derivatives for a given function are always equal if both cross partials exist and they are continuous. This means

$$Z_{xy} = Z_{yx} = \frac{\partial Z_y}{\partial x} = \frac{\partial Z_x}{\partial y}$$

Example

1. Given the demand function for the two commodities Q_1 and Q_2

$$Q_1 = P_1^{-1.7} P_2^{0.8} \quad \text{And} \quad Q_2 = P_1^{0.5} P_2^{-0.2}$$

Find the first and second order partial derivatives of the function and determine whether the two goods are complementary or substitute. Show that whether young's theorem is satisfied or not.

$$\begin{aligned} \frac{\partial Q_1}{\partial P_1} &= -1.7 P_1^{-2.7} P_2^{0.8} & \frac{\partial Q_2}{\partial P_1} &= 0.5 P_1^{-0.5} P_2^{-0.2} \\ \frac{\partial Q_1}{\partial P_2} &= 0.8 P_1^{-1.7} P_2^{-0.2} & \frac{\partial Q_2}{\partial P_2} &= -0.2 P_1^{0.5} P_2^{-1.2} \end{aligned}$$

From these first order partial derivatives we have observed that an increase in the price of commodity Q_1 results in an increase in the quantity demanded of commodity Q_2 . Therefore, Q_1 and Q_2 are substitute goods.

The second order direct partial derivatives are

$$\begin{aligned} \frac{\partial^2 Q_1}{\partial P_1^2} &= (-1.7)(-2.7) P_1^{-3.7} P_2^{0.8} \\ &= 4.59 P_1^{-3.7} P_2^{0.8} \\ \frac{\partial^2 Q_2}{\partial P_2^2} &= 0.24 P_1^{0.5} P_2^{-2.2} \end{aligned}$$

The second cross partial derivatives are

$$\begin{aligned}\frac{\partial}{\partial p_2} \left(\frac{\partial Q_1}{\partial p_1} \right) &= (0.8) (-1.7) P_1^{-2.7} P_2^{-0.2} \\ &= -1.36 P_1^{-2.7} P_2^{-0.2} \\ \frac{\partial}{\partial P_1} \left(\frac{\partial Q_1}{\partial P_2} \right) &= (-1.7)(0.8) P_1^{-2.7} P_2^{-0.2} \\ &= -1.36 P_1^{-2.7} P_2^{-0.2} \\ \frac{\partial}{\partial P_1} \left(\frac{\partial Q_2}{\partial p_2} \right) &= (-0.2) (0.5) P_1^{-0.5} P_2^{-1.2} \\ &= -0.1 P_1^{-0.5} P_2^{-1.2} \\ \frac{\partial}{\partial P_2} \left(\frac{\partial Q_2}{\partial p_1} \right) &= (-0.2) (0.5) P_1^{-0.5} P_2^{-1.2} \\ &= -0.1 P_1^{-0.5} P_2^{-1.2}\end{aligned}$$

2. Given the production function $Q = 6K + 0.3 K^2L + 1.2L^2$ derive the four second order partial derivatives and interpret their meaning.

The two first order partial derivatives are

$$\frac{\partial Q}{\partial K} = 6 + 0.6 KL \quad \text{and} \quad \frac{\partial Q}{\partial L} = 0.3 K^2 + 2.4 L$$

They represent the marginal product function of the two inputs.

The four second order partial derivatives are

$$\text{i) } \frac{\partial^2 Q}{\partial K^2} = 0.6L$$

This represents the rate of change of marginal product of capital function with respect to capital. It indicates that marginal product of capital increases as labor increases.

$$\text{ii) } \frac{\partial^2 Q}{\partial L^2} = 2.4$$

It represents the slope of marginal product of labor function which is independent of capital.

$$\text{iii) } \frac{\partial^2 Q}{\partial L \partial K} = 0.6K$$

This tells us that MP_L increases if K is increased, for any given value of L .

$$\text{iv) } \frac{\partial^2 Q}{\partial L \partial K} = 0.6K$$

This tells us that MP_K increases if L is increased, for any given value of L .

2.3.3 Multivariate chain Rule

As we know many economic models involve composite functions. Thus differentiation of such functions requires the application of chain rule.

- Chain rule for functions of one variable

Given $y = f(x)$ and $x = g(t)$, then $\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}$

- Chain rule for multivariate function

Suppose $z = f(x, y)$, $x = f(t)$ and $y = f(t)$, in this case we should use the total derivative.

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \left(\frac{dx}{dt} \right) + \frac{\partial z}{\partial y} \left(\frac{dy}{dt} \right)$$

Given the function $y = f(x, w)$ and $x = g(w)$ to get this total derivative, let us initially determine the total differential $dy = f_x dx + f_w dw$. If we divide both sides of this equation by dw , we obtain

$$\frac{dy}{dw} = f_x \frac{dx}{dw} + f_w \frac{dw}{dw}$$

$$\frac{dy}{dw} = f_x \frac{dx}{dw} + f_w$$

Where $f_x \frac{dx}{dw}$ measures the indirect effect of w on y and f_w shows the direct effect of w on y .

Example

1. Suppose that the relationship between revenue R , output Q produced and sold each week is given by

$$R = 400Q - Q^2$$

In addition, suppose that Q is a function of t , i.e. time period, $Q = f(t)$

$$\text{Thus } \frac{dR}{dt} = \frac{dR}{dQ} \frac{dQ}{dt}$$

$$= (400 - 2Q) \frac{dQ}{dt}$$

When $Q = 30$ and that the Management is considering to increase production by five units per week,

$$\frac{dR}{dt} = [400 - 2(30)]5$$

$$= (400 - 60)5$$

$$\frac{dR}{dt} = (340)5 = 1700$$

Interpretation

The management decision to increase production by five units per week increases total revenue by 1700 units.

2. Given the utility function $U = x^2 + x y + y^2$, $x = t^3 + 1$ and $y = t - t^3$

$$\begin{aligned} \frac{du}{dt} &= \frac{\partial u}{\partial x} \left(\frac{dx}{dt} \right) + \frac{\partial u}{\partial y} \left(\frac{dy}{dt} \right) \\ &= (2x+y) (3t^2) + (y+2y) (1-3t^2) \end{aligned}$$

2.3.4 Homogenous Functions and Euler's Theorem

A function $y = f(x_1, x_2, \dots, x_n)$ is referred to as homogenous of degree r if and only if

$$f(tx_1, tx_2, \dots, tx_n) = t^r f(x_1, x_2, \dots, x_n), \text{ where } t \text{ is any parameter.}$$

Multiplying all the independent variables by a factor t will multiply the value of the function by the factor t^r .

Example

1. If $f(x, y) = 3x^3 + 5x^2y + 2y^3$, then

$$f(tx, ty) = 3(tx)^3 + 5(tx)^2(ty) + 2(ty)^3$$

$$= 3t^3x^3 + 5t^2x^2y + 2t^3y^3$$

$$= 3t^3x^3 + 5t^3x^2y + 2t^3y^3$$

$$= t^3(3x^3 + 5x^2y + 2y^3)$$

Therefore, the function is homogenous of degree 3.

2. Consider the Cobb- Douglas production function

$$Q = bL^\alpha K^{1-\alpha}$$

$$f(Lt, kt) = b(Lt)^\alpha (kt)^{1-\alpha}$$

$$= b t^\alpha L^\alpha k^{1-\alpha} t^{1-\alpha}$$

$$= b t^{\alpha+1-\alpha} L^\alpha k^{1-\alpha}$$

$$= b t L^\alpha k^{1-\alpha}$$

$$= t b L^\alpha k^{1-\alpha}$$

Thus, the production function is homogenous of degree 1.

Given the function $f(tx_1, tx_2, tx_3, \dots, tx_n) = t^r f(x_1, x_2, x_3, \dots, x_n)$

If $r = 1$, a function exhibits constant returns to scale.

If $r > 1$, a function exhibits increasing returns to scale.

If $r < 1$, a function exhibits decreasing returns to scale.

Euler's Theorem 1.

If the function $y = f(x_1, x_2, x_3)$ is homogenous of degree r , then according to the Euler's theorem

$$\frac{\partial f}{\partial x_1} x_1 + \frac{\partial f}{\partial x_2} x_2 + \frac{\partial f}{\partial x_3} x_3 = r f(x_1, x_2, x_3)$$

Example

Find the degree of homogeneity of the function

$$f(x, y) = x^4 + x^2y^2 \text{ Using Euler's theorem}$$

$$f_x x + f_y y = r(x^4 + x^2y^2)$$

$$(4x^3 + 2xy^2)x + (2x^2y)y = r(x^4 + x^2y^2)$$

$$= 4x^4 + 2x^2y^2 + 2x^2y^2$$

$$= 4x^4 + 4x^2y^2$$

$$= 4(x^4 + x^2 y^2)$$

Thus, the function is homogenous of degree 4.

Euler's theorem 2

If a function $y = f(x_1, x_2, \dots, x_n)$ is homogenous of degree r , then according to this theorem the first partials $f_{x_1}, f_{x_2}, f_{x_3}$ are homogenous of degree $(r - 1)$.

Example

1. Given the function $f(x, y) = x^4 + x^2 y^2$,

$$\begin{aligned} f_x &= 4x^3 + 2xy^2 & f_y &= 2x^2 y \\ f_x(tx, ty) &= 4(tx)^3 + 2(tx)(ty)^2 \\ &= 4t^3 x^3 + 2t^3 xy^2 \\ &= t^3(4x^3 + 2xy^2) \end{aligned}$$

Thus, the function f_x is homogenous of degree 3.

$$\begin{aligned} f_y(tx, ty) &= 2(tx)^2 (ty) \\ &= 2t^2 x^2 ty \\ &= 2t^3 x^2 y = t^3(2x^2 y) \end{aligned}$$

This means, f_y is homogenous of degree 3.

2. Given the Cobb - Douglas production function

$$f(L, K) = 60 L^{3/4} K^{1/4}$$

Find the degree of homogeneity of f_L and f_K using Euler's theorem.

The first partial derivatives are

$$\begin{aligned} f_L &= 60 \left(\frac{3}{4}\right) L^{-1/4} K^{1/4} & f_K &= \frac{1}{4} (60) L^{3/4} K^{-3/4} \\ f_L &= \frac{45K^{1/4}}{L^{1/4}} & f_K &= \frac{15L^{3/4}}{K^{3/4}} \end{aligned}$$

$$\begin{aligned} \text{Thus, } f_L(tL, tK) &= 45(tL)^{-1/4} (tK)^{1/4} \\ &= 45 L^{-1/4} t^{-1/4} K^{1/4} t^{1/4} \\ &= 45(t)^{(1/4 - 1/4)} L^{-1/4} K^{1/4} \\ &= t^0 (45 L^{-1/4} K^{1/4}) \end{aligned}$$

$$\begin{aligned} \text{And } f_K(tL, tK) &= 15(tL)^{3/4} (tK)^{-3/4} \\ &= 15 L^{3/4} t^{3/4} K^{-3/4} t^{-3/4} \\ &= 15(t)^{(3/4 - 3/4)} L^{3/4} K^{-3/4} \end{aligned}$$

$$= t^0 (15 L^{\frac{3}{4}} K^{\frac{-3}{4}})$$

What we have seen from this is both of f_L and f_K is homogenous of degree 0. In other words, the original Cobb-Douglas production is homogenous of degree 1.

2.3.5. Total Differential and Implicit Differentiation

Given the function $y = f(x)$, the differential is $dy = f'(x)dx$. The concept of differential can be extended to functions with two or more independent variables.

Given the utility function, $U = f(x_1, x_2)$, assuming that U is continuous and differentiable

$$U_{x_1} = \frac{\partial U}{\partial x_1} = \text{Marginal utility of } x_1$$

$$U_{x_2} = \frac{\partial U}{\partial x_2} = \text{Marginal utility of } x_2$$

Change in U which is resulted from small change in x_1 is $\frac{\partial U}{\partial x_1} dx_1$ and from small change in x_2 is

$$\frac{\partial U}{\partial x_2} dx_2.$$

Thus the total change in utility is

$$dU = \frac{\partial U}{\partial x_1} dx_1 + \frac{\partial U}{\partial x_2} dx_2$$

If the two variables x and y are related by the implicit function $f(x, y) = 0$, the total differential of the function is

$$d0 = f_x dx + f_y dy$$

$$0 = f_x dx + f_y dy$$

If we rearrange this equation, we get

$$0 = f_x dx + f_y dy$$

$$\frac{dy}{dx} = -\frac{f_x}{f_y}$$

As you remember in your microeconomic studies, the theory of production states that the slope of the isoquant represents the marginal rate of technical substitution (MRTS) between two inputs. In this case, the use of total differential can help us to show the reality that MRTS between the two inputs is equal to the ratio of the marginal products of the two inputs. If the producer use capital

(K) and labor (L) in the production process, $MRTS_{LK} = \frac{dK}{dL}$

Given the production function $Q = f(L, K)$, the total differential is

$$dQ = \frac{\partial Q}{\partial K} dK + \frac{\partial Q}{\partial L} dL$$

Once the isoquant is determined, there is no change in output along the same isoquant. This means change in Q is zero. Therefore,

$$\begin{aligned}\frac{\partial Q}{\partial K} dK + \frac{\partial Q}{\partial L} dL &= 0 \\ \frac{\partial Q}{\partial K} dK &= -\frac{\partial Q}{\partial L} dL \\ \frac{dK}{dL} &= \frac{-\partial Q / \partial L}{\partial Q / \partial K} = \frac{-MP_L}{MP_K}\end{aligned}$$

$\text{MRTS}_{Lk} = \frac{-MP_L}{MP_K}$
